

Stratos Retail Intelligence

Industry Context

Stratos Retail Intelligence is a data analytics company that helps retail organizations analyze sales transactions, understand customer purchasing behavior, and evaluate product performance across markets. Using structured datasets containing **sales records, customer information, and product catalogs**, the company enables organizations to track revenue performance, customer segments, and product demand using modern analytics tools.

Students must analyze the provided datasets (**Sales.csv, Customers.csv, Products.csv**) using Excel, SQL, Python, and Power BI.

Excel

Question 1

Create a **Pivot Table** showing **Total Revenue per Product Category and Brand**.

- Merge **Sales.csv** and **Products.csv** using *ProductKey*
- Create a column:
Revenue = Quantity × Unit Price USD

Question 2

Calculate the **Average Revenue per Customer**.

- Merge **Sales.csv** and **Customers.csv** using *CustomerKey*
- Compute:

Total Revenue per Customer
Total Orders per Customer

Formula:

Average Order Value = Total Revenue / Total Orders

SQL

Question 3

Write a SQL query to calculate **Total Revenue per Product Category**.

- Join **Sales and Products** tables
- Revenue formula:

Revenue = Quantity × Unit Price USD

Return:

Category
Total Revenue

Question 4

Write a SQL query to identify **Top 10 Cities with Highest Revenue**.

- Join **Sales and Customers** tables
- Calculate:

Total Revenue per City

Sort by **Revenue descending**.

Python

Question 5

Explore the dataset using pandas.

Perform:

```
df.head()
df.tail()
df.columns
df.isnull().sum()
```

Then merge **Sales, Customers, and Products** datasets and create a new column:

Revenue = Quantity × Unit Price USD

Question 6

Create visualizations using **matplotlib / seaborn**:

- Line chart showing **Revenue trend over time**
- Histogram showing **Revenue distribution**
- Swarm plot showing **Revenue across Product Categories**

Power BI

Question 7

Create a **Sales Dashboard** showing:

- Total Revenue KPI
- Revenue by Product Category (Bar Chart)
- Revenue Trend over Time (Line Chart)

Question 8

Create a visual report showing:

- Revenue by Country
- Revenue by Brand
- Revenue distribution across Product Categories

Use the datasets **Sales, Customers, and Products** to build the visuals.